Notes: Sequeira (AER, 2016)

Corruption, Trade Costs, and Gains from Tariff Liberalization: Evidence from Southern Africa

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1 One-sentence summary

This note summarizes the paper, which uses the SADC tariff phaseout between South Africa and Mozambique, aggregate COMTRADE trade flows, a panel survey of Mozambican firms, and an audit dataset of more than 1,000 import shipments to show that nominal tariff liberalization generated very small import-volume elasticities because pervasive customs corruption had already reduced effective tariff costs, while the reform still lowered bribe payments and improved the accuracy of reported trade flows.

2 Big picture

2.1 Main question

The paper asks a simple but important question: **when tariffs fall sharply, do imports actually respond?** Standard trade models predict that lower tariffs should reduce trade costs and increase imported quantities. Yet, in some developing-country environments, firms may not actually pay the official tariff because customs officials can sell tariff evasion through bribes. In that case, the official tariff is not the relevant marginal trade cost.

2.2 Core answer

The paper's answer is that **official tariff reductions can have weak effects on trade flows when corruption already lowers effective tariffs**. The Mozambican tariff reduction was large in statutory terms: treated goods went from positive tariffs to zero, and the average nominal tariff rate fell by roughly 30 percent. But import quantities responded only weakly. The paper argues that this weak response is not primarily because firms face no potential suppliers, nor because of pure transport barriers, but because many firms were already avoiding tariffs through bribes.

2.3 Key mechanism

The mechanism is:

1. Before liberalization, official tariffs were high for some products.
2. Customs officials could offer importers tariff evasion in exchange for bribes.
3. Bribes were frequent and sizable in absolute terms, but small relative to the tariff duties avoided.
4. Therefore, the *effective* tariff faced by firms was much lower than the statutory tariff.
5. When the statutory tariff was removed, the actual trade cost faced by many firms changed little.
6. As a result, import quantities and firm sourcing decisions responded only weakly.

2.4 Main empirical punchline

The paper finds an estimated trade elasticity of roughly **0.1**, close to the low end of time-series estimates in the trade literature. But this low elasticity is not interpreted as a deep structural preference parameter. Instead, it is interpreted as an object contaminated by corruption-induced measurement of trade costs: the official tariff is not the cost firms actually pay.

3 Incremental contribution of the paper

3.1 Contribution to trade elasticity estimation

The paper contributes to the literature on trade elasticities by estimating the import response to a genuine policy-driven change in tariffs. Many trade elasticity estimates rely on cross-sectional variation or on tariff variation that may be endogenous to trade flows, lobbying, or industry structure. Here, the tariff reductions were predetermined under the SADC Trade Protocol, with the key schedule agreed long before the main empirical period.

The incremental contribution is not only the estimate itself, but the interpretation: **very low elasticities may arise when official trade-cost measures differ from actual costs paid by firms**. This is particularly relevant for countries where customs corruption is pervasive.

3.2 Contribution to corruption measurement

The paper also contributes to corruption measurement. The standard approach in trade-related corruption work is to infer tariff evasion from trade gaps: differences between exporter-reported exports and importer-reported imports. Sequeira complements this indirect measure with primary data on actual bribe payments during the import process.

This is important because trade gaps only capture some forms of corruption, especially under-reporting or misclassification. The audit data identify:

- whether a bribe was paid;
- how much was paid;
- who received it;
- why it was paid;
- whether the bribe was collusive or coercive;
- whether bribes shifted from tariff evasion to other forms after liberalization.

3.3 Contribution to understanding gains from tariff liberalization

The paper shows that applying the Arkolakis-Costinot-Rodriguez-Clare gains-from-trade formula mechanically with an extremely low estimated elasticity can yield implausibly large welfare gains. The key lesson is that welfare calculations based on official tariff changes may be misleading if official tariffs are not the relevant trade costs.

3.4 Contribution to policy design

The paper also clarifies that tariff liberalization may still matter even if trade volumes do not respond strongly. It can:

- reduce transfers from firms to customs officials;
- reduce incentives to misreport import quantities;
- improve the reliability of trade statistics;
- reduce the role of corrupt intermediaries in public service delivery.

Thus, tariff reform can have governance benefits even when standard import-volume responses are weak.

4 Institutional setting and source of variation

4.1 The SADC tariff phaseout

Mozambique joined the Southern African Development Community in 1992 and committed to the SADC Trade Protocol in 1996. The agreement required the phaseout of tariffs on imports from South Africa by 2015. The most important reduction for this paper occurred in 2007-2008, when the average nominal tariff rate fell by around 5 percentage points, approximately a 30 percent decline.

A central identification advantage is that the tariff phaseout schedule was set in the mid-1990s, before the study period and before the later transformation of the Mozambican and South African economies. The paper argues that this makes the tariff changes plausibly unrelated to contemporary product-level lobbying or product-specific demand shocks.

4.2 Treatment and control products

The empirical design compares products that experienced tariff reductions with products that did not. The paper shows that, prior to the 2008 tariff change, treatment and comparison products were broadly similar in terms of:

- import shares by quantity;
- import shares by value;
- unit values;
- shipment values per ton;
- number of containers;
- containerization status.

The main exception is whether the product has an internationally set reference price, which matters for tariff evasion because reference prices make misreporting harder.

4.3 Why the setting is useful

This setting is useful because it combines:

1. a predetermined tariff reform;
2. variation across products in whether tariffs changed;
3. aggregate trade-flow data before and after the reform;
4. firm-level survey data before and after the reform;
5. direct shipment-level data on corruption before and after the reform.

Together, these data allow the paper to connect macro trade elasticities to micro-level corruption and firm behavior.

5 Data

5.1 Aggregate trade-flow data

The paper uses COMTRADE data for imports from South Africa into Mozambique between 2006 and 2014. Products are measured at the HS six-digit level. Tariff rates come from the Mozambican

Customs Authority. The aggregate analysis uses exporter-reported data when possible, because exporters have less incentive to misreport tariff-relevant import quantities.

5.2 Firm survey data

The paper uses a two-wave survey of 190 formal Mozambican firms in 2006 and 2010. These firms operated in the hinterland of the trade corridor connecting South Africa and Mozambique. The survey measures:

- whether a firm imports its main input;
- the percentage of the main input imported;
- the tariff grouping of the main input;
- firm size;
- ownership structure;
- age of establishment;
- ethnicity and gender of the firm manager.

The key limitation is that this panel covers incumbent firms, not all potential entrants or exiters.

5.3 Audit data on bribe payments

The paper's most distinctive data source is an audit study of more than 1,000 shipments imported into Mozambique between 2007 and 2013. A random sample of clearing agents tracked shipments through the import process and reported:

- shipment value and tonnage;
- product category;
- timing of arrival and clearance;
- whether a bribe was paid;
- amount of the bribe;
- recipient of the bribe;
- reason for the bribe;
- whether the bribe was collusive or coercive.

The audit data are especially valuable because they identify the actual corrupt transaction, rather than only inferring corruption from discrepancies in trade statistics.

6 Empirical designs and equations

6.1 Aggregate trade elasticity specification

The first empirical object is the elasticity of imports with respect to tariffs. The main aggregate specification is:

$$\log ImportShare_{it} = \alpha_0 + \alpha_1 \log TariffRate_{it} + \mu_i + \gamma_t + \varepsilon_{it}. \tag{1}$$

Here, $ImportShare_{it}$ is imports of product i from South Africa relative to imports from the rest of the world into Mozambique. The coefficient α_1 is the trade elasticity of interest.

The paper estimates this model using fixed effects, first differences, long differences, and an instrumental variables strategy. The IV specification uses lagged tariff rates and baseline 2006 tariffs to address measurement error in the tariff variable.

6.2 Firm-level extensive and intensive margins

For the extensive margin, the dependent variable captures whether the firm stopped importing, remained in the same sourcing status, or started importing. The specification is:

$$D.ImportStatus_k = \alpha_0 + \alpha_1 D.Tariff_k + \alpha_3 X_k + \omega_j + \varepsilon_k. \tag{2}$$

For the intensive margin, the dependent variable is the change in the percentage of the main input imported:

$$D.PctgImportedInput_k = \alpha_0 + \alpha_1 D.Tariff_k + \alpha_3 X_k + \omega_j + \varepsilon_k. \tag{3}$$

The firm-level results test whether incumbent firms changed import status or import intensity in response to tariff reductions.

6.3 Trade-gap difference-in-differences

The trade-gap analysis uses the difference between quantities, values, or unit values reported by South African exporters and Mozambican importers:

$$\log TradeGap_{it} = \gamma_1(Treated_i \times POST_t) + \beta POST_t + \gamma_2 Treated_i + \beta_2 BaselineTariff_i + \varepsilon_{it}. \tag{4}$$

A fall in the quantity trade gap after tariff removal suggests that importers become more truthful in reporting quantities once the incentive to evade tariffs is reduced.

6.4 Bribe-payment difference-in-differences

The audit-data specification is:

$$y_{it} = \gamma_1(TariffChangeCategory_i \times POST_t) + \mu POST_t \tag{5}$$

$$+ \gamma_2 TariffChangeCategory_i + \beta_2 BaselineTariff_i + \Gamma_i + \rho_i + \omega_t + \delta_i + \varepsilon_{it}. \tag{6}$$

The dependent variable y_{it} is either an indicator for paying a bribe, the log amount of bribe paid, or another bribe outcome. The key coefficient is γ_1, which measures whether products exposed to tariff liberalization experienced a differential change in corruption after 2008.

7 Main results

7.1 Official tariffs fell, but trade volumes barely moved

The aggregate results show very small import elasticities. The preferred IV estimate is around -0.097, and the fixed-effect and difference specifications are even closer to zero. This means a large tariff cut produced only a small increase in import volumes. The result is surprising because:

- the policy shock was permanent;

- firms knew the phaseout schedule in advance;
- the analysis covers several years after the reform;
- lower uncertainty should itself increase trade.

The unit-value regressions show that firms did not respond by upgrading quality instead of expanding quantity.

7.2 Firm-level data confirm weak behavioral adjustment

The firm panel suggests that tariff reductions did not meaningfully change either:

- the extensive margin: whether firms started or stopped importing; or
- the intensive margin: how much of the main input firms imported.

This supports the aggregate finding that the statutory tariff reduction was inframarginal for incumbent firms.

7.3 Corruption before the reform was pervasive

Before tariff liberalization, around 80 percent of audited shipments involved a bribe payment. Most bribes went to customs officials and were primarily linked to tariff evasion. Bribes were large relative to port charges but small relative to the tariff duties avoided. This fits the Tullock paradox: the payment for the illicit transaction is far smaller than the rent the importer obtains.

7.4 Tariff removal reduced bribe payments

The tariff reform reduced both the probability and the amount of bribe payments:

- Products that experienced tariff reductions were about 30 percent less likely to pay a bribe.
- Conditional on paying a bribe, bribe amounts declined substantially.
- Bribes as shares of shipment value and shipment tonnage also fell.

This supports the interpretation that tariffs created rents that customs officials and importers shared through tariff evasion.

7.5 Tariff removal improved reporting accuracy

The trade-gap evidence shows that tariff removal reduced the gap in reported quantities between South African exporters and Mozambican importers. More directly, the paper shows that Mozambican importers reported higher volumes after tariff removal, whereas South African exporters did not show comparable reporting changes. This is strong evidence that the earlier trade gap reflected strategic misreporting by Mozambican importers.

7.6 Displacement effects existed but were limited

After tariff liberalization, some corruption shifted from collusive tariff evasion to more coercive forms of bribe extraction, such as bribes linked to delays, queue jumping, or documentation issues. However, the paper argues that these displacement effects were limited. They reduced the total decline in corruption by less than 10 percent and did not generate a large new trade-cost burden.

7.7 Gains-from-trade calculations become misleading

Using the ACR formula with the estimated elasticity of about 0.1 implies an implausibly high real-income gain of 87 percent. The paper interprets this as a warning: if the elasticity is estimated using statutory tariffs that firms do not actually pay, then the elasticity is not the correct primitive for welfare calculations.

8 Tables and figures from the paper

8.1 Tables 1–2: Comparability of Trade Patterns and Product Characteristics; Tariff Liberalization and Import Volumes

Read: Table 1 establishes that treatment and control products were broadly comparable before the tariff change. Table 2 shows the first main result: import volumes respond very weakly to official tariff reductions, with the preferred IV elasticity around -0.097.

Economic message: The tariff reform is a credible policy shock, but the import-volume response is much smaller than standard trade models would lead us to expect.

Table 1—Comparability of Trade Patterns and Product Characteristics across Treatment and Control Products, Prior to the 2008 Tariff Change

	Treatment products		Comparison products		Difference
	Mean	SD	Mean	SD	t-test/χ^2 p-value
Panel A. Patterns of trade ($N = 4{,}660$)					
Share of imports (quantities)	38.1	492.5	241.2	5244.4	0.20
Share of imports (value)	22.2	237.2	33.84	341.2	0.19
Unit value	4.82	53.7	6.21	83.3	0.51
Panel B. Product characteristics ($N = 265$)					
Shipment value per ton (USD)	44,027	179,869	410,508	2,959,621	0.25
Number of containers per shipment	7.99	2.606	7.4	3.173	0.10
Bulk cargo (non-containerized)					0.94
Rauch product classification					0.01

Notes: In panel A, the share of imports corresponds to the share of imports into Mozambique from South Africa relative to imports from the Rest of the World. Treatment products are all products affected by the tariff change and comparison products are all products that did not experience a change in tariffs. p-values are for t-tests of equality of means of continuous variables, assuming unequal variances. In panel B, the data correspond to the sample of shipments captured in the audit study conducted by the author in 2007.

Sources: United Nations COMTRADE database, Mozambican customs tariff code, and audit study conducted by the author

(a) Table 1: Balance between treatment and comparison products.

Table 2—Tariff Liberalization and Import Volumes, 2006–2014: Aggregate Import Flows

	log share import volumes			
	Fixed effects (1)	First differences (2)	Long differences (3)	Instrumental variable (4)
Panel A. 2 SLS Estimate				
log tariff rate	−0.016			−0.097
	(0.027)			(0.050)
Δ log tariff rate		−0.010	−0.076	
		(0.019)	(0.018)	
Panel B. First stage dep. var. log tariff rate				
Lagged log tariff rate (one period)				0.841
				(0.042)
Lagged log tariff rate (two periods)				−0.085
				(0.011)
Baseline tariff rate 2006				−0.040
				(0.002)
Kleibergen-Paap Wald F-statistic				207.09
Observations	21,520	16,353	13,022	15,326
Mean of dependent variable	1.094	1.051	1.055	1.130

Notes: log share import volumes corresponds to the quantity of imports from South African as a share of the total quantity of imports from the rest of the world into Mozambique, in logarithmic form. Robust standard errors clustered at the level of the product's four-digit HS code.

(b) Table 2: Tariff liberalization and import volumes.

8.2 Tables 3–4: Tariff Liberalization and Unit Values; Firm-Level Import Responses

Read: Table 3 tests whether tariff cuts induced quality upgrading, using unit values as a proxy. Table 4 uses firm-level survey data to test extensive and intensive margins of importing.

Economic message: Firms did not meaningfully adjust quantities, quality, or sourcing status after the tariff cut. The aggregate low elasticity is visible at the firm level.

Table 3—Tariff Liberalization and Unit Value of Imports, 2006–2014: Aggregate Import Flows

	log share unit value			
	Fixed effects (1)	First differences (2)	Long differences (3)	Instrumental variable (4)
Panel A. 2 SLS Estimate				
log tariff rate	0.001			0.002
	(0.011)			(0.013)
Δ log tariff rate		−0.008	0.080	
		(0.012)	(0.011)	
Panel B. First stage dep. var. log tariff rate				
Lagged log tariff rate (one period)				0.841
				(0.042)
Lagged log tariff rate (two periods)				−0.085
				(0.011)
Baseline tariff rate 2006				−0.040
				(0.002)
Kleibergen-Paap Wald *F*-statistic				207.09
Observations	21,520	16,353	12,977	15,326
Mean of dependent variable	0.706	0.673	0.664	0.667

Notes: log share import volumes corresponds to the unit value of South African imports as a share of the unit value of imports from the rest of the world into Mozambique, in logarithmic form. Robust standard errors clustered at the level of the product's four-digit HS code.

Sources: United Nations COMTRADE database and Mozambican customs tariff code.

(a) Table 3: Tariff liberalization and unit values.

Table 4—Tariff Liberalization and Import Volumes, 2006–2010: Firm-Level Data

	Extensive margin change in import status Ordered probit			Intensive margin change in pctg of imp. input Ordinary least squares		
	(1)	(2)	(3)	(4)	(5)	(6)
Δ log tariff rate	−0.113	−0.106	−0.091	−0.738	0.689	1.681
	(0.079)	(0.136)	(0.160)	(2.306)	(3.927)	(4.936)
Firm size		0.539	0.534		10.127	9.355
		(0.229)	(0.238)		(7.212)	(7.154)
Ethnicity of owner		0.188	0.187		7.388	5.681
		(0.187)	(0.178)		(7.430)	(6.839)
Foreign firm		0.456	0.436		15.155	14.250
		(0.329)	(0.269)		(14.330)	(12.665)
Age of establishment		−0.016	−0.018		−0.400	−0.431
		(0.009)	(0.009)		(0.251)	(0.242)
log baseline tariff 2006		0.046	0.036		7.059	5.170
		(0.170)	(0.174)		(4.933)	(5.223)
Controls						
Industry fixed effects	No	No	Yes	No	No	Yes
Observations	160	117	117	160	117	117
p-value of joint significance of FE		0.000			0.000	

Notes: Change in import status corresponds to −1 if the firm stopped importing, 0 if the firm did not change its import status, and 1 if the firm started to import.

Source: Enterprise Surveys conducted by the author in 2006 and 2010

(b) Table 4: Firm-level importing responses.

8.3 Tables 5–6: Trade Gaps and Tariff Levels; Summary Statistics on Bribe Payments

Read: Table 5 uses trade gaps to show that tariffs are associated with quantity misreporting. Table 6 documents the prevalence and type of bribe payments before and after the reform.

Economic message: The primary channel of evasion is quantity under-reporting, and before liberalization most bribes were paid to customs officials for tariff evasion.

Table 5—Trade Gaps and Tariff Levels, 2006–2014

	log trade gap					
	Quantity (1)	Value (2)	Unit value (3)	Quantity (4)	Value (5)	Unit value (6)
log tariff	0.201	0.055	−0.013			
	(0.042)	(0.035)	(0.010)			
Treated products × POST				−0.493	−0.083	0.022
				(0.097)	(0.077)	(0.031)
Treated products				0.308	−0.092	0.104
				(0.243)	(0.219)	(0.087)
POST				0.385	0.118	0.394
				(0.086)	(0.068)	(0.028)
log baseline tariff				0.245	0.271	−0.051
				(0.114)	(0.098)	(0.043)
Observations	21,884	21,884	21,861	21,884	21,884	21,861
Mean of dependent variable	0.273	0.213	2.176	0.273	0.213	2.178
R^2	0.187	0.165	0.675	0.170	0.155	0.422

Notes: Trade gap corresponds to the difference between quantities, price, and unit values reported by exporters in South Africa and the equivalent figures reported by importers in Mozambique. The treated products indicator equals 1 if products experienced a change in tariff in 2008 and 0 otherwise. POST corresponds to the periods after the 2008 tariff change.

(a) Table 5: Trade gaps and tariff levels.

Table 6—Summary Statistics: Bribe Payments

	Pre-tariff change	Post-tariff change	
	2007	2008	2011–2012
Probability of paying a bribe (percent)	80	26	16
Avg bribe amount per ton (Metical 2007, CPI adjusted)	2,164	280	494
	(7,800)	(963)	(2,746)
Primary bribe recipient	Customs (97%)	Customs (84%)	Customs (72%)
Primary reason for bribe payment	Tariff evasion (61%)	Congestion (59%)	Congestion (38%)
Ratio of bribe amount to tariff duties saved [0–1]*	0.07	0.028	0.008
	(0.13)	(0.09)	(0.02)
Average clearing time for all shipments (days)	2.6	2.6	2.6
	(2.2)	(1.3)	(3.6)
Average clearing time with the payment of a bribe (days)	2.6	2.2	2.4
	(2.3)	(1.0)	(3.1)
Average clearing time without the payment of a bribe (days)	1.9	2.7	2.6
	(1.2)	(1.4)	(3.7)
Average clearing time with bribe payment for tariff evasion (days)	2.7	2.4	2.4
	(2.4)	(1.0)	(1.8)

Notes: Average clearing times moved in tandem with increases in the overall volume of cargo handled at the port between 2007 and 2011. Total volumes increased by 13 percent in 2008 and 18 percent in 2011. Note that in 2009, the port of Maputo was still functioning at 30 percent of capacity so it was capable of handling the observed increase in volumes without substantially increasing congestion.

*Conditional on the bribe being paid for tariff evasion.

Source: Audit study conducted by the author

(b) Table 6: Summary statistics on bribe payments.

8.4 Tables 7–8: Bribes by Shipper and Product Characteristics; Probability of Paying a Bribe

Read: Table 7 breaks bribe patterns down by shipper and product characteristics. Table 8 estimates the effect of tariff liberalization on the probability of paying a bribe.

Economic message: Bribes fell across many shipment types, and treated products experienced a large decline in the probability of paying a bribe after tariffs were removed.

Table 7—Bribes Before and After the Tariff Change: By Shipper and Product Characteristics

	Pre-tariff change	Post-tariff change	Difference *p*-value
Panel A. Probability of paying a bribe (percent)			
Large firm	96	16	0.000
Medium to small firm	67	18	0.000
Agricultural product	13	12	0.739
Differentiated product	77	18	0.000
Pre-inspected shipment	68	10	0.000
Panel B. Amount of bribe paid per ton (Mtn, CPI adjusted)			
Large firm	3,373	150	0.004
	(1,419)	(75)	
Medium to small firm	3,882	503	0.000
	(1,711)	(85)	
Agricultural product	1,404	615	0.144
	(922)	(143)	
Differentiated product	2,062	537	0.000
	(623)	(90)	
Pre-inspected shipment	2,597	661	0.000
	1,136	130	

Notes: Panel B reports standard errors in parenthesis. The *t*-tests for equality of means reported in panel B assume unequal variances. Large firms are defined as having more than 100 employees. Differentiated products correspond to products that lack a reference international price (as defined in Rauch 1999). The pre-inspected shipment dummy variable denotes whether a shipment was inspected at origin.

Source: Audit study conducted by the author

(a) Table 7: Bribes before and after the tariff change.

Table 8—Difference-in-Differences: Determinants of the Probability of Paying a Bribe

	Probability of paying a bribe [0–1] linear probability model			
	(1)	(2)	(3)	(4)
Tariff change category × POST	−0.429	−0.296		
	(0.131)	(0.120)		
Tariff change category	0.448	0.357		
	(0.111)	(0.099)		
Tariff reduction × POST			−0.025	−0.021
			(0.008)	(0.007)
Tariff reduction			0.024	0.022
			(0.005)	(0.009)
POST	−0.089	−0.555	−0.111	−0.686
	(0.106)	(0.203)	(0.116)	(0.241)
Differentiated product	0.065	0.018	0.032	−0.076
	(0.078)	(0.102)	(0.071)	(0.109)
Agricultural product	0.026	−0.221	0.046	0.041
	(0.030)	(0.096)	(0.029)	(0.030)
Pre-shipment inspection	−0.010	0.061	0.003	0.087
	(0.010)	(0.061)	(0.020)	(0.07)
Perishable product	−0.047	0.260	−0.052	0.137
	(0.067)	(0.109)	(0.064)	(0.124)
Large firm	0.058	0.161	0.066	0.172
	(0.047)	(0.055)	(0.051)	(0.066)
log shipment value per ton	0.014	−0.035	0.017	−0.034
	(0.008)	(0.011)	(0.008)	(0.013)
Controls				
Clearing agent fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Product four-digit HS code	Yes	Yes	Yes	Yes
Terminal	Yes	Yes	Yes	Yes
Day of the week arrival	Yes	Yes	Yes	Yes
Product from South Africa	Yes	Yes	Yes	Yes
Baseline tariff	Yes	Yes	Yes	Yes
Covariates × POST	No	Yes	No	Yes
Observations	1,084	1,084	1,084	1,084
F-statistic	218.28	3,308.08	313.90	604.80
R^2	0.36	0.39	0.35	0.37

Notes: The dependent variable equals 1 if a bribe was paid and 0 otherwise. The tariff change category indicator equals 1 if the product experienced a tariff change in 2008 and 0 otherwise. The tariff reduction variable corresponds to the percentage point reduction in tariffs experienced by each product in 2008. Robust standard errors clustered at the level of the four-digit HS code.

Sources: Audit study conducted by the author and Mozambican customs tariff code

(b) Table 8: Probability of paying a bribe.

8.5 Tables 9–10: Amount of Bribe Paid; Bribes as a Share of Shipment Value

Read: Table 9 shows that tariff liberalization reduced the amount of bribe paid, conditional on paying a bribe. Table 10 scales bribes by shipment value.

Economic message: The reform reduced not only the frequency of corruption, but also the magnitude of bribes relative to economic activity.

Table 9—Difference-in-Differences: Determinants of the Amount of Bribe Paid

	log bribe amount paid							
	Ordinary least squares				Hurdle model			
	(1)	(2)	(3)	(4)	logit (5)	Negative binomial (6)	logit (7)	Negative binomial (8)
Tariff change category × POST	−3.748	−2.928			−30.735	−0.079		
	(1.075)	(0.944)			(1.995)	(0.459)		
Tariff change category	3.632	3.156			30.704	−0.916		
	(0.953)	(0.803)			(1.898)	(0.436)		
Tariff reduction × POST			−0.225	−0.191			−2.996	−0.089
			(0.058)	(0.064)			(0.174)	(0.031)
Tariff reduction			0.200	0.191			2.969	−0.042
			(0.042)	(0.0478)			(0.171)	(0.0260)
POST	−0.678	−3.449	−0.864	−4.652	−0.392	−0.633	−0.371	−0.426
	(0.867)	(1.818)	(0.944)	(2.152)	(0.639)	(0.179)	(0.634)	(0.200)
Differentiated product	0.545	−0.121	0.303	−0.925	−0.0450	0.188	−0.104	0.304
	(0.648)	(0.849)	(0.603)	(0.876)	(0.660)	(0.423)	(0.643)	(0.427)
Agricultural product	0.161	−1.968	0.343	0.337	0.356	0.583	0.327	0.229
	(0.285)	(0.931)	(0.265)	(0.243)	(0.365)	(0.563)	(0.355)	(0.494)
Pre-shipment inspection	−0.227	0.376	−0.137	0.641	−0.122	−0.550	−0.102	−0.595
	(0.208)	(0.628)	(0.197)	(0.712)	(0.215)	(0.182)	(0.207)	(0.189)
Perishable product	−0.084	3.400	−0.119	2.299	−0.551	0.311	−0.711	0.768
	(0.616)	(0.845)	(0.586)	(0.949)	(1.147)	(0.787)	(1.167)	(0.748)
Large firm	0.600	1.593	0.662	1.708	1.137	0.270	1.198	0.277
	(0.389)	(0.486)	(0.431)	(0.585)	(0.610)	(0.391)	(0.618)	(0.393)
log shipment value per ton	0.130	−0.221	0.152	−0.217	0.160	−0.037	0.158	−0.035
	(0.074)	(0.079)	(0.073)	(0.095)	(0.088)	(0.079)	(0.088)	(0.077)
Controls								
Clearing agent fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product four-digit HS code	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Terminal	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of the week arrival	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product from South Africa	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline tariff	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Covariates × POST	No	Yes	No	Yes	No	No	No	No
Observations	1,084	1,084	1,084	1,084	1,084	1,084	1,084	1,084
Mean of dependent variable	1.486	1.486	1.486	1.486	1.486	1.486	1.486	1.486
F-statistic	265.38	3,937.63	422.36	188.25				
R^2	0.35	0.38	0.35	0.37				
log pseudo-likelihood					−2,243.93		−2,243.58	

Notes: Columns 1 through 4 denote estimates for an ordinary least squares model, with the dependent variable representing the logarithmic form of the amount of bribe paid, conditional on paying a bribe. Columns 5 through 8 fit a hurdle model with a truncated negative binomial, in levels, to ensure convergence. The negative binomial is preferred over a truncated Poisson model due to over-dispersion in the bribe amount variable. Tariff change category equals 1 if the product experienced a tariff change in 2008 and 0 otherwise. Tariff reduction corresponds to the magnitude of the tariff reduction experienced by each product in 2008. Robust standard errors clustered at the level of the four-digit HS code.

Sources: Audit study conducted by the author and Mozambican customs tariff code

(a) Table 9: Amount of bribe paid.

Table 10—Difference-in-Differences: Bribes as a Share of Shipment Value

	log bribe amount paid as a share of shipment value OLS			
	(1)	(2)	(3)	(4)
Tariff change category × POST	−0.010	−0.010		
	(0.003)	(0.004)		
Tariff change category	0.010	0.010		
	(0.003)	(0.004)		
Tariff reduction × POST			−0.001	−0.001
			(0.0580)	(0.0642)
Tariff reduction			0.001	0.001
			(0.000)	(0.000)
POST	−0.002	−0.001	−0.002	−0.003
	(0.001)	(0.011)	(0.001)	(0.011)
Differentiated product	0.010	0.007	0.010	0.005
	(0.005)	(0.007)	(0.004)	(0.006)
Agricultural product	−0.001	−0.004	−0.000	−0.000
	(0.001)	(0.004)	(0.001)	(0.001)
Pre-shipment inspection	−0.000	0.002	0.000	0.002
	(0.001)	(0.005)	(0.001)	(0.005)
Perishable product	0.001	0.004	0.001	0.002
	(0.002)	(0.004)	(0.002)	(0.004)
Large firm	−0.003	−0.004	−0.003	−0.003
	(0.002)	(0.004)	(0.002)	(0.004)
log shipment tonnage	−0.000	−0.001	−0.000	−0.002
	(0.000)	(0.001)	(0.000)	(0.001)
Controls				
Clearing agent fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Product four-digit HS code	Yes	Yes	Yes	Yes
Terminal	Yes	Yes	Yes	Yes
Day of the week arrival	Yes	Yes	Yes	Yes
Product from South Africa	Yes	Yes	Yes	Yes
Baseline tariff	Yes	Yes	Yes	Yes
Covariates × POST	No	Yes	No	Yes
Observations	1,084	1,084	1,084	1,084
Mean of dependent variable	0.001	0.001	0.001	0.001
F-statistic	5.93	3.61	9.77	6.80
R^2	0.435	0.461	0.431	0.452

Notes: Columns 1 through 4 denote estimates from an ordinary least squares model, with the dependent variable representing the logarithmic form $[\log(x+1)]$ of the amount of bribe paid as a share of the value of the shipment, conditional on paying a bribe. Tariff change category equals 1 if the product experienced a tariff change in 2008 and 0 otherwise. Tariff reduction corresponds to the magnitude of the tariff reduction experienced by each product in 2008. Robust standard errors clustered at the level of the four-digit HS code.

Sources: Audit study conducted by the author and Mozambican customs tariff code

(b) Table 10: Bribes as a share of shipment value.

8.6 Tables 11–12: Bribes as a Share of Shipment Tonnage; Excluding Agricultural Products

Read: Table 11 scales bribes by shipment tonnage. Table 12 shows that results are robust when agricultural products, which are more likely to face nontariff barriers, are excluded.

Economic message: The decline in corruption is not just an artifact of shipment size or value, and low elasticities remain when excluding a sector especially exposed to nontariff barriers.

Table 11—Difference-in-Differences: Bribe Amounts Paid as a Share of Shipment Tonnage

	log bribe amount paid as a share of shipment tonnage			
	OLS			
	(1)	(2)	(3)	(4)
Tariff change category × POST	−1.914	−1.597		
	(0.504)	(0.462)		
Tariff change category	1.875	1.680		
	(0.465)	(0.427)		
Tariff reduction × POST			−0.089	−0.084
			(0.026)	(0.027)
Tariff reduction			0.084	0.087
			(0.022)	(0.023)
POST	−0.274	−3.003	−0.420	−3.713
	(0.318)	(1.237)	(0.393)	(1.527)
Differentiated product	0.182	−0.079	0.085	−0.375
	(0.258)	(0.439)	(0.234)	(0.420)
Agricultural product	0.117	−0.701	0.198	0.129
	(0.128)	(0.441)	(0.112)	(0.102)
Pre-shipment inspection	−0.104	0.299	−0.056	0.491
	(0.082)	(0.424)	(0.077)	(0.471)
Perishable product	0.410	2.110	0.389	1.607
	(0.324)	(0.381)	(0.307)	(0.513)
Large firm	−0.202	−0.253	−0.162	−0.155
	(0.241)	(0.470)	(0.264)	(0.500)
log shipment value	−0.001	−0.173	0.001	−0.189
	(0.031)	(0.072)	(0.032)	(0.083)
Controls				
Clearing agent fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Product four-digit HS code	Yes	Yes	Yes	Yes
Terminal	Yes	Yes	Yes	Yes
Day of the week arrival	Yes	Yes	Yes	Yes
Product from South Africa	Yes	Yes	Yes	Yes
Baseline tariff	Yes	Yes	Yes	Yes
Covariates × POST	No	Yes	No	Yes
Observations	1,084	1,084	1,084	1,084
Mean of dependent variable	0.451	0.451	0.451	0.451
F-statistic	23.66	36.26	21.38	18.58
R^2	0.249	0.279	0.228	0.262

Notes: Columns 1 through 4 denote estimates for an ordinary least squares model, with the dependent variable representing the logarithmic form of the amount of bribe paid as a share of the value of the shipment, conditional on paying a bribe. Tariff change category equals 1 if the product experienced a tariff change in 2008 and 0 otherwise. Tariff reduction corresponds to the magnitude of the tariff reduction experienced by each product in 2008. Robust standard errors clustered at the level of the four-digit HS code.

Source: Audit study conducted by the author and Mozambican customs tariff code

(a) Table 11: Bribes as a share of shipment tonnage.

Table 12—Tariff Liberalization and Unit Value of Imports, 2006–2014: Excluding Agricultural Products

	log share unit value			
	Fixed effects (1)	First differences (2)	Long differences (3)	Instrumental variable (4)
				A. 2 SLS Estimate
log tariff rate	−0.060			−0.101
	(0.036)			(0.046)
Δ log tariff rate		−0.013	−0.061	
		(0.023)	(0.021)	
				B. First stage dep. var. log tariff rate
Lagged log tariff rate (one period)				0.861
				(0.0626)
Lagged log tariff rate (two periods)				−0.076
				(0.020)
Baseline tariff rate 2006				−0.057
				(0.003)
Kleibergen-Paap Wald F-statistic				136.211
Observations	13,292	10,178	8,483	9,434
Mean of dependent variable	0.948	0.911	0.924	0.958

Notes: log share import volumes corresponds to the unit value of South African imports (in kgs) as a share of the total quantity of imports from the rest of the world into Mozambique, in logarithmic form. Robust standard errors clustered at the level of the product's six-digit harmonization code.

Sources: United Nations COMTRADE database and Mozambican customs tariff code

(b) Table 12: Excluding agricultural products.

8.7 Tables 13–14: Parallel Trend Hypothesis; Firm Characteristics and Selection

Read: Table 13 tests the parallel trend assumption using a placebo 2007 reform. Table 14 checks whether attrition in the firm panel creates selection problems.

Economic message: The identifying assumptions are supported: pre-trends do not suggest differential movements before treatment, and firm attrition does not appear to be strongly selective.

Table 13—The Parallel Trend Hypothesis: Aggregate Trade Flow Data

	Quantity trade gap	Value trade gap	Unit value trade gap
Panel A. Trade gaps with placebo 2007 tariff liberalization			
Tariff change category × POST 2007	−0.038	−0.006	−0.073
	(0.134)	(0.111)	(0.053)
Tariff change category	0.031	−0.159	0.702
	(0.227)	(0.181)	(0.142)
POST 2007	0.079	0.088	0.094
	(0.118)	(0.098)	(0.045)
log baseline tariff	0.204	0.121	−0.088
	(0.084)	(0.067)	(0.058)
Observations	4,655	5,423	4,640
R^2	0.002	0.012	0.039
Panel B. Trade gaps with actual 2008 tariff liberalization			
Treated products × POST 2008	−0.493	−0.083	0.022
	(0.097)	(0.077)	(0.031)
Treated products	0.308	−0.092	0.104
	(0.243)	(0.219)	(0.087)
log baseline tariff	0.245	0.271	−0.051
	(0.114)	(0.098)	(0.043)
POST	0.385	0.118	0.394
	(0.086)	(0.068)	(0.028)
Observations	21,884	21,884	21,861
R^2	0.170	0.155	0.422

Notes: Tariff change category equals 1 if the product changed tariff category in 2008 and 0 otherwise. Log baseline tariff rate represents the product's tariff rate in 2006, in logarithmic form. Standard errors clustered at the four-digit HS code.

Sources: United Nations COMTRADE database and Mozambican customs tariff code

(a) Table 13: Parallel trend placebo and actual reform.

Table 14—Firm Characteristics: Selection

	Attrition panel			Attrition covariates		
	t-test *p*-value	Kolmogorov-Smirnov *p*-value	χ^2 *p*-value	*t*-test *p*-value	Kolmogorov-Smirnov *p*-value	χ^2 *p*-value
Firm age	0.86	0.34		0.90	0.26	
Sales in 2006	0.12	0.37		0.40	0.77	
Number of workers in 2006	0.51	0.36		0.18	0.92	
Percentage of imported inputs	0.73	0.85		0.24	0.45	
Percentage female ownership	0.55	0.99		0.01	0.13	
Firm exports in 2006			0.29			0.33
Ethnicity of firm manager			0.31			0.15

Notes: *p*-values for *t*-tests of equality of means of continuous variables firm age, sales in 2006, number of workers in 2006, and percentage of imported inputs (assuming unequal variances). Exact Kolmogorov-Smirnov *p*-values for a non-parametric test of the equality of distributions of continuous variables and *p*-values for a χ^2 test for equality of categorical variables.

Sources: Enterprise Surveys conducted by the author in 2006 and 2010

(b) Table 14: Firm selection and attrition checks.

8.8 Tables 15–16: Tariff Changes and Misreporting of Trade Flows; Monitoring Experiment

Read: Table 15 examines whether reported trade flows change after tariffs fall. Table 16 reports the monitoring experiment used to validate bribe reporting.

Economic message: Importer-side reporting becomes more accurate after tariff removal, while the monitoring experiment suggests self-administered bribe reports are informative but that observation can change behavior.

Table 15—Tariff Changes and Misreporting of Trade Flows

	log volume (in kgs)		log value	
	Mozambique (1)	South Africa (2)	Mozambique (3)	South Africa (4)
Tariff change category × POST	0.456	−0.048	0.132	0.039
	(0.092)	(0.092)	(0.075)	(0.079)
Tariff change category	−0.775	−0.390	0.714	0.619
	(0.395)	(0.429)	(0.395)	(0.388)
POST	0.136	0.526	0.463	0.594
	(0.081)	(0.081)	(0.066)	(0.069)
Controls				
Year fixed effects	Yes	Yes	Yes	Yes
Product four-digit HS code fixed effects	Yes	Yes	Yes	Yes
Observations	21,909	21,909	21,909	21,909
Mean of dependent variable	8.619	8.896	10.500	10.718
R^2	0.01	0.01	0.01	0.02

Notes: Tariff change category equals 1 if the product experienced a tariff change in 2008 and 0 otherwise. POST indicator equals 1 for years 2008–2014 and 0 for years 2006–2007.

Sources: United Nations COMTRADE database and Mozambican customs tariff code

(a) Table 15: Tariff changes and misreporting.

Table 16—Monitoring Experiment

	Bribe paid				log bribe amount paid			
	Linear probability model				OLS			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Monitored shipment	−0.100	−0.101	−0.200	−0.190	−0.951	−0.895	−1.647	−1.418
	(0.035)	(0.025)	(0.057)	(0.080)	(0.295)	(0.179)	(0.476)	(0.639)
Monitored shipment			0.044	0.037			0.308	0.214
× log tariff level			(0.017)	(0.027)			(0.138)	(0.216)
Controls								
Industry fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
Clearing agent	No	Yes	No	Yes	No	Yes	No	Yes
Product four-digit HS code	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Terminal	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month arrival	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Differentiated product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Agricultural product	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Perishable	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-inspected shipment	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Large client firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of the week arrival	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
log value per ton	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
log tariff level	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bulk (non-containerized) shipments	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	919	919	919	919	919	919	919	919
F-test	5.31	15.49	24.68	24.04	6.71	16.82	18.02	30.76
R^2	0.05	0.09	0.05	0.09	0.04	0.08	0.04	0.08

Notes: In columns 1 through 4 the dependent variable equals 1 if a bribe was paid and 0 otherwise, while in columns 5 through 8 the dependent variable corresponds to the amount of bribe paid in logarithmic form $[\ln(x+1)]$. Columns 1 through 4 fit linear probability models while columns 5 through 8 fit standard ordinary least squares. Robust standard errors are clustered at the four-digit HS code for each product.

Sources: Audit study conducted by the author

(b) Table 16: Monitoring experiment.

8.9 Tables 17–18: Shipment Characteristics; Shift from Collusive to Coercive Corruption

Read: Table 17 checks whether shipment composition changed across the audit waves. Table 18 examines whether corruption shifted from collusive bribes to coercive bribes.

Economic message: Shipment samples remain broadly comparable, and although some displacement toward coercive corruption occurred, it was not large enough to undo the overall fall in corruption.

Table 17—Summary Statistics: Shipment Characteristics

Shipment characteristics	Difference Pre-post tariff change (p-values)		
Panel A	All products	Treatment products	Control products
Average shipment value per ton	0.28	0.97	0.27
Percentage bulk cargo	0.83	0.93	0.69
Perishable product	0.47	0.01	0.04
Percentage of differentiated goods	0.71	0.003	0.16
Panel B	High-High	High-Low	Low-Low
Average shipment value per ton	0.19	0.58	0.36
Percentage bulk cargo	0.01	0.20	0.53
Perishable product	0.001	0.000	0.56
Percentage of differentiated goods	0.03	0.01	0.06

Notes: The high-high grouping represents products that remained in the high tariff category; the high-low grouping represents products that went from high to low tariffs and the low-low grouping represents products that remained in a low tariff category throughout the period under analysis. Bulk represents non-containerized cargo. For the continuous variable average shipment value per ton, t-test for equality of means, accounting for unequal variances. For all other binary variables, the p-value corresponds to a χ^2 test.

Source: Audit study conducted by the author

(a) Table 17: Shipment characteristics.

Table 18—Shift from Collusive to Coercive Corruption

	Probability of paying a coercive bribe Linear probability model				Shift collusive to coercive Ordered probit	
	(1)	(2)	(3)	(4)	(5)	(6)
Tariff change category × POST	0.442	0.318			0.834	
	(0.163)	(0.149)			(0.475)	
Tariff change category	0.083	0.097			0.094	
	(0.137)	(0.104)			(0.471)	
Tariff reduction × POST			0.029	0.020		0.018
			(0.010)	(0.012)		(0.030)
Tariff reduction			0.014	0.007		0.041
			(0.007)	(0.006)		(0.029)
POST	0.295	0.174	0.231	0.127	0.864	1.140
	(0.089)	(0.461)	(0.111)	(0.422)	(1.801)	(1.706)
Differentiated product	0.040	0.041	−0.021	0.031	0.350	0.114
	(0.110)	(0.124)	(0.139)	(0.12)	(0.579)	(0.554)
Perishable product	−0.194	0.177	−0.236	0.091	0.239	0.046
	(0.145)	(0.192)	(0.160)	(0.187)	(0.778)	(0.776)
Firm size	−0.066	−0.105	−0.059	−0.096	−0.986	−0.871
	(0.064)	(0.076)	(0.058)	(0.079)	(0.382)	(0.385)
log shipment value per ton	0.025	−0.017	0.0167	−0.019	0.0350743	0.018
	(0.019)	(0.030)	(0.015)	(0.029)	(0.116)	(0.110)
Controls						
Day of the week arrival	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Clearing agent	Yes	Yes	Yes	Yes	Yes	Yes
Pre-shipment inspection	Yes	Yes	Yes	Yes	Yes	Yes
Log baseline tariff	Yes	Yes	Yes	Yes	Yes	Yes
Covariates × post	No	Yes	No	Yes	Yes	Yes
Observations	228	220	228	220	1,084	1,084
F-statistic	30.69	27.98	18.54	15.88		
Wald statistic					1,156	1,102.08
log pseudo-likelihood					−591.316	−598.758
R^2	0.32	0.32	0.29	0.29		

Notes: Columns 1 through 4 represent a linear probability model, where the dependent variable equals 1 if a coercive bribe was paid and 0 if a collusive bribe was paid instead. Columns 5 and 6 represent an ordered probit in which the dependent variable equals −1 if a collusive bribe was paid, 0 if no bribe was paid and 1 if a coercive bribe was paid. All columns represent the standard difference-in-differences specification. In columns 1, 2, and 5 the treatment variable equals 1 if the product falls under a tariff category that experienced a tariff reduction in 2008 and 0 otherwise; in columns 3, 4, and 8 the treatment variable captures the percentage point tariff reduction experienced by each product. Robust standard errors are correlated at the level of the four-digit HS code.

Source: Audit study conducted by the author

(b) Table 18: Shift from collusive to coercive corruption.

8.10 Table 19: Tariff Liberalization and Bribe Dispersion

Read: Table 19 examines whether tariff liberalization changed the dispersion of bribe payments.

Economic message: Treated products experienced a decline in bribe dispersion, suggesting that tariff removal lowered not only the level of corrupt payments but also some of their uncertainty.

Table 19—Tariff Liberalization and Bribe Dispersion

Panel A. Test of equality of means	Before tariff change	After tariff change	Difference	p-value
Standard deviation of bribes per ton (all products)	2,021 (236)	1,692 (50)	329	0.17
Standard deviation of bribes per ton (treatment products)	2,633 (448)	903 (140)	−1,531	0.00
Standard deviation of bribes per ton (bribes paid to customs for tariff evasion)	1,652 (446)	1,059 (449)	593	0.35
Standard deviation of bribes per ton (bribes paid to customs by treatment products)	2,448 (450)	1,316 (392)	−1,133	0.06

	SD of bribe amount paid per ton, OLS	
Panel B. Dependent variable	(1)	(2)
Tariff change category × post	−1,976	−3,742
	(1,599)	(1,835)
Tariff change category	1,999	3,567
	(1,479)	(1,754)
Post	320	−5,893
	(315)	(4,739)
Controls		
Clearing agent fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Terminal	Yes	Yes
Day of the week arrival	Yes	Yes
Product from South Africa	Yes	Yes
log baseline tariff	Yes	Yes
Covariates × post	Yes	Yes
Observations	1,072	1,072
Mean of dependent variable	2.622	2.622
R^2	0.49	0.53

Notes: Standard deviation of bribes per ton calculated at the level of the four-digit HS code. Differences (panel A) and robust standard errors (panel B) significant at 1 percent, 5 percent, and 10 percent.

Source: Audit study conducted by the author

Table 19: Tariff liberalization and bribe dispersion.

9 How to interpret the paper

9.1 Why the estimated elasticity is low

The low elasticity is not interpreted as evidence that Mozambican firms are intrinsically insensitive to prices. Rather, the low elasticity reflects the fact that official tariffs were not the true marginal cost. If firms already reduced their tariff burden through bribes, the statutory tariff reduction does not generate a large cost shock.

9.2 Why corruption can reduce measured tariff effects

A simple way to see the logic is:

$$\text{Effective tariff cost} = \text{official tariff paid after evasion} + \text{bribe paid.} \tag{7}$$

If the official tariff is high but most of it can be evaded in exchange for a small bribe, then the effective tariff cost may be low. Tariff liberalization removes both the official tariff and some bribe payments, but if the pre-reform effective cost was already low, the net fall in trade costs can be small.

9.3 Why tariff reform still matters

Even with low import-volume responses, tariff reform matters because it changes distributional and institutional outcomes. It reduces rents for customs officials, lowers the probability of corrupt transactions, reduces misreporting incentives, and improves statistical accuracy. These are governance effects, not necessarily immediate trade-volume effects.

10 Limitations and caveats

10.1 Incumbent-firm limitation

The firm-level evidence focuses on incumbent firms. It cannot fully assess whether new firms entered importing or whether potential entrants were affected by the tariff reform. If tariff cuts matter primarily through entry of new importers, the firm survey could miss part of the response.

10.2 External validity

The results are especially relevant for environments where customs corruption is pervasive and where bribes are tied to tariff evasion. They may not generalize to countries with strong enforcement, low corruption, or digitalized customs processes.

10.3 Measurement of bribes

Although the audit study is unusually rich, bribe measurement remains difficult. The monitoring experiment shows that observation can reduce reported bribe payments, which means that the act of measuring corruption can change corruption behavior. The paper addresses this directly, but measurement remains an important caveat.

10.4 Interpreting welfare gains

The paper does not claim that tariff liberalization produced no welfare benefits. Rather, it argues that conventional welfare calculations can be distorted if the estimated elasticity is based on official tariffs that do not measure actual trade costs.

11 Short summary

- The paper studies a large tariff reduction under the SADC agreement between South Africa and Mozambique.
- Aggregate trade-flow data show very low import elasticities, around 0.1.
- Firm survey data show little change in import status or import intensity among incumbent firms.
- Trade gaps show that tariff evasion operated mainly through quantity misreporting.
- Shipment-level audit data show pervasive pre-reform corruption: around 80 percent of shipments involved bribes.
- Most pre-reform bribes were paid to customs officials for tariff evasion.
- Tariff liberalization reduced both the probability and size of bribe payments.
- Some corruption shifted from collusive tariff evasion to coercive bribes, but the displacement was limited.
- The paper's central lesson is that tariff liberalization may produce small trade-volume responses when corruption has already reduced effective tariffs.

12 Conclusion

Sequeira (2016) shows that the effectiveness of tariff liberalization depends on the institutional environment in which tariffs are enforced. In Mozambique, official tariffs were only part of the real trade-cost structure because corruption allowed firms to evade duties at a relatively low bribe cost. As a result, large statutory tariff cuts generated weak import responses. But the reform still reduced corruption, improved reporting incentives, and lowered transfers from firms to officials.

The paper's broader contribution is to connect trade policy, corruption, and measurement. It warns that the official policy variable may not be the economic variable firms face. In settings with weak enforcement, estimates of trade elasticities and gains from trade can be seriously distorted unless researchers measure both formal trade costs and the informal institutions that mediate them.